

Newman-Penrose Formalism Calculations

Stephani et al., Equation (12.26)

Metric

$$ds^2 = -a^2 dt \otimes dt - a^2 e^x dt \otimes dy - a^2 e^x dy \otimes dt - \frac{1}{2} a^2 e^{(2x)} dy \otimes dy + a^2 dz \otimes dz + a^2 dx \otimes dx$$

Coordinates

$$(x^\mu) = (t, y, z, x)$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2} \sqrt{2} a \right) dt + \left(\frac{1}{2} \sqrt{2} a e^x \right) dy + \left(\frac{1}{2} \sqrt{2} a \right) dz$$

$$n = \left(\frac{1}{2} \sqrt{2} a \right) dt + \left(\frac{1}{2} \sqrt{2} a e^x \right) dy + \left(-\frac{1}{2} \sqrt{2} a \right) dz$$

$$m = \left(\frac{1}{2} i a e^x \right) dy + \left(\frac{1}{2} \sqrt{2} a \right) dx$$

$$\bar{m} = \left(-\frac{1}{2} i a e^x \right) dy + \left(\frac{1}{2} \sqrt{2} a \right) dx$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}}{2a} \right) dt + \left(\frac{\sqrt{2}}{2a} \right) dz$$

$$n = \left(-\frac{\sqrt{2}}{2a}\right) dt + \left(-\frac{\sqrt{2}}{2a}\right) dz$$

$$m = \left(-\frac{i}{a}\right) dt + \left(\frac{i e^{(-x)}}{a}\right) dy + \left(\frac{\sqrt{2}}{2a}\right) dx$$

$$\bar{m} = \left(\frac{i}{a}\right) dt + \left(-\frac{i e^{(-x)}}{a}\right) dy + \left(\frac{\sqrt{2}}{2a}\right) dx$$

Directional Derivatives

$$D = -\frac{\sqrt{2}}{2a} \partial_t + \frac{\sqrt{2}}{2a} \partial_z$$

$$\Delta = -\frac{\sqrt{2}}{2a} \partial_t - \frac{\sqrt{2}}{2a} \partial_z$$

$$\delta = -\frac{i}{a} \partial_t + \frac{i e^{(-x)}}{a} \partial_y + \frac{\sqrt{2}}{2a} \partial_x$$

$$\bar{\delta} = \frac{i}{a} \partial_t - \frac{i e^{(-x)}}{a} \partial_y + \frac{\sqrt{2}}{2a} \partial_x$$

Spin Coefficients

$$\rho = -\frac{i}{2a}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{i}{2a}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\sqrt{2}}{4a}$$

$$\beta = \frac{\sqrt{2}}{4a}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{i}{4a}$$

$$\gamma = -\frac{i}{4a}$$

Ricci Tensor Components

$$\Phi_{00} = \frac{1}{4a^2}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{1}{8\,a^2}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{1}{4\,a^2}$$

$$\Lambda = -\frac{1}{24\,a^2}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{1}{6\,a^2}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type " D "